

GDL 2026 Conference Program

28-30 September 2026 | Day 1 registration: 08:15-09:00

Day 1: 09:00-17:00 | Day 2: 08:00-16:00 | Day 3: 09:00-18:00

Day 1 | Monday, 28 September 2026

Venue: Nanyang Executive Centre Auditorium

MORNING | 08:15-13:10

TIME	PROGRAM
08:15-09:00	Registration 45 min
09:00-09:10	Opening remarks 10 min
09:10-10:10	KEYNOTE 60 MIN Equivariant Systems Theory Robert Mahony Australian National University Chair: Juan-Pablo Ortega
10:10-10:30	Coffee break 20 min
10:30-12:10	SESSION 1 PART I OF II 6 TALKS 150 MIN TOTAL Manifold, Kernel and Topological Learning Chair: Juan-Pablo Ortega
10:30-10:55	A Geometer's View on the Manifold Hypothesis Wilderich Tuschmann Karlsruhe Institute of Technology
10:55-11:20	MMD-Regularized Unbalanced Optimal Transport Piyushi Manupriya Indian Institute of Technology Hyderabad
11:20-11:45	SPD Matrix Learning for Brain-Computer Interfaces and NeuroAI Ce Ju The Chinese University of Hong Kong, Shenzhen
11:45-12:10	Invariant kernels on symmetric spaces Cyrus Mostajeran Nanyang Technological University
12:10-13:10	Lunch 60 min

Day 1 | Monday, 28 September 2026

Venue: Nanyang Executive Centre Auditorium

AFTERNOON | 13:10-17:00

TIME	PROGRAM
13:10-14:00	SESSION 1 PART II OF II 6 TALKS 150 MIN TOTAL Manifold, Kernel and Topological Learning Chair: Frank Nielsen
13:10-13:35	Kernel Learning of PDE Solution Operators Jianyu Hu <i>Nanyang Technological University</i>
13:35-14:00	kNN: From a Topological Data Analysis (TDA) Perspective to a Topological Combinatorics Approach Minh Le Quang <i>Ho Chi Minh City Open University</i>
14:00-14:50	SESSION 2 PART I OF II 4 TALKS 100 MIN TOTAL Dynamical Learning Architectures Chair: Frank Nielsen
14:00-14:25	Rank-one fluctuations in averaging-learning dynamics: agreement geometry without a ground truth Tushar Vaidya <i>Nanyang Technological University</i>
14:25-14:50	The analogy between ResNets and ODEs misses an expectation Lénaïc Chizat <i>National University of Singapore</i>
14:50-15:25	Coffee break 35 min
15:25-16:15	SESSION 2 PART II OF II 4 TALKS 100 MIN TOTAL Dynamical Learning Architectures Chair: Cyrus Mostajeran
15:25-15:50	State-space and state-affine systems for learning from temporal data Florian Rossmannek <i>Nanyang Technological University</i>
15:50-16:15	Reservoir computing through the lens of observer theory and dominant manifolds Anastasia Bizyaeva <i>Cornell University</i>
16:15-17:00	KEYNOTE 45 MIN Chair: Cyrus Mostajeran Geometric adjoint sensitivity analysis with applications to neural ODEs and PDE constrained optimization Melvin Leok <i>University of California San Diego</i>

Day 2 | Tuesday, 29 September 2026

Venue: Nanyang Executive Centre Function Hall 1 (Cypress 1)

MORNING | 08:00-12:15

TIME	PROGRAM
08:00-08:45	KEYNOTE 45 MIN Chair: Frank Nielsen Information-Geometric Port-Hamiltonian Reinforcement Learning Chandrajit Bajaj <i>The University of Texas at Austin</i>
08:45-09:35	SESSION 3 PART I OF II 5 TALKS 125 MIN TOTAL Chair: Frank Nielsen Symmetry, Equivariance and Geometric Control
08:45-09:10	Directions at Infinity: from statistical depth to hyperbolic curvature Yangdi Jiang <i>Nanyang Technological University</i>
09:10-09:35	Robust subspace tracking using Grassmannian geometry Ravi N. Banavar <i>Indian Institute of Technology Bombay</i>
09:35-09:55	Coffee break 20 min
09:55-10:40	KEYNOTE 45 MIN Chair: Francois Gay-Balmaz An invariant theoretic perspective on learning path dependences in finance and technology Josef Teichmann <i>ETH Zurich</i>
10:40-12:15	SESSION 4 PART I OF II 7 TALKS 195 MIN TOTAL Chair: Francois Gay-Balmaz (Non)Dissipative Thermodynamics and Physics-Informed Machine Learning
10:40-11:25	SPECIAL SESSION OPENING TALK 45 MIN Souriau's Dissipative Lie Groups Thermodynamics-Informed Machine Learning, Lee's Conformal Hamiltonian Dynamics & Kubo's Fluctuation-Dissipation extension Frédéric Barbaresco <i>THALES</i>
11:25-11:50	General Properties of the Thermodynamic Metrics for Generalized Souriau Partition Functions on CV manifolds Pietro Fré <i>University of Turin; INFN Turin</i>
11:50-12:15	A Universal Obstruction to the Samuelson Condition for Tangent Lagrangian 2-Webs Yasuhiro Kurokawa <i>Shibaura Institute of Technology</i>

Day 2 | Tuesday, 29 September 2026

Venue: Nanyang Executive Centre Function Hall 1 (Cypress 1)

AFTERNOON | 12:15-16:00

TIME	PROGRAM
12:15-13:15	Lunch 60 min
13:15-14:55	SESSION 4 PART II OF II 7 TALKS 195 MIN TOTAL Chair: Frédéric Barbaresco (Non)Dissipative Thermodynamics and Physics-Informed Machine Learning
13:15-13:40	A Geometric Perspective on Asymmetric Relaxation Dynamics Alessandro Bravetti <i>University of Camerino</i>
13:40-14:05	Beyond PINNs: From Natural-Gradient Geometry to Hybrid Galerkin Methods Nilo Schwencke <i>Inria; ENS de Lyon; CNRS</i>
14:05-14:30	Lie-Poisson Neural Network with control and dissipation for multi-rotor flight mechanics Meng Wu <i>Nanyang Technological University</i>
14:30-14:55	Structured preserving Lagrangian neural methods for Vlasov equations with turbulence Emmanuel Franck <i>Inria; University of Strasbourg; CNRS</i>
14:55-15:15	Coffee break 20 min
15:15-16:00	KEYNOTE 45 MIN Chair: Jérémie Houssineau Learning, approximation and control Qianxiao Li <i>National University of Singapore</i>

Day 3 | Wednesday, 30 September 2026

Venue: School of Physical and Mathematical Sciences (SPMS) Lecture Theatre 2 (SPMS-03-03)

MORNING | 09:00-13:05

TIME	PROGRAM
09:00-09:45	KEYNOTE 45 MIN Chair: Juan-Pablo Ortega Geometric Integration and Its Applications to Optimization and Machine Learning David Martín de Diego <i>Instituto de Ciencias Matemáticas (ICMAT-CSIC)</i>
09:45-11:00	SESSION 3 PART II OF II 5 TALKS 125 MIN TOTAL Chair: Juan-Pablo Ortega Symmetry, Equivariance and Geometric Control
09:45-10:10	Beyond Symmetry: Efficient, Generalizable Learning for Weakly Invariant Systems Jake Welde <i>Cornell University</i>
10:10-10:35	Synchronous Models and Fundamental Systems in Observer Design Pieter van Goor <i>University of Sydney</i>
10:35-11:00	Generalized Homogeneity and Its Applications Andrey Polyakov <i>Inria; University of Lille; CNRS</i>
11:00-11:25	Coffee break 25 min
11:25-13:05	SESSION 5 PART I OF II 6 TALKS 150 MIN TOTAL Chair: Kelin Xia Information Geometry and Variational Inference
11:25-11:50	Statistical Potentials and Inverse Problems in Information Geometry: A Bi-Tensorial Approach Marco Pacelli <i>Scuola Superiore Meridionale; University of Naples Federico II</i>
11:50-12:15	New faces of Bregman divergences Frank Nielsen <i>Sony Computer Science Laboratories</i>
12:15-12:40	Bayesian Learning at Scale: Relaxations, Duality and Algorithms Thomas Möllenhoff <i>RIKEN</i>
12:40-13:05	Maxitive Donsker-Varadhan Formulation for Possibilistic Variational Inference Jérémie Houssineau <i>Nanyang Technological University; University of Warwick</i>

Day 3 | Wednesday, 30 September 2026

Venue: School of Physical and Mathematical Sciences (SPMS) Lecture Theatre 2 (SPMS-03-03)

AFTERNOON | 13:05-18:00

TIME	PROGRAM
13:05-14:05	Lunch 60 min
14:05-14:55	SESSION 5 PART II OF II 6 TALKS 150 MIN TOTAL Information Geometry and Variational Inference Chair: Florian Rossmannek
14:05-14:30	The Geometry of Conditional Distributions Nathael da Costa University of Tübingen
14:30-14:55	Information flow, its geometry and information inequalities in stochastic flows: From uncertainty quantification in data-driven state estimation of non-autonomous dynamics to machine learning Michal Branicki Great Bay University
14:55-15:45	SESSION 6 PART I OF II 6 TALKS 150 MIN TOTAL Stochastic Learning, Optimization and Gradient Flows Chair: Florian Rossmannek
14:55-15:20	From stochastic dynamics to agentic learning equations, and toward AI scaling Lancelot Da Costa Max Planck Institute for Intelligent Systems
15:20-15:45	Policy Gradient Descent for Stochastic Control Zhenjie Ren Université Évre Paris-Saclay
15:45-16:10	Coffee break 25 min
16:10-17:50	SESSION 6 PART II OF II 6 TALKS 150 MIN TOTAL Stochastic Learning, Optimization and Gradient Flows Chair: Cyrus Mostajeran
16:10-16:35	New quantum algorithms based on Riemannian optimization Marek Gluza Nanyang Quantum Solutions; Nanyang Technological University
16:35-17:00	Spectral Optimization on Manifolds: Geometry and Robustness Bamdev Mishra Microsoft India
17:00-17:25	Deep learning over Riemannian manifolds Ziheng Chen University of Trento; Max Planck Institute for Intelligent Systems
17:25-17:50	Shape Regression and Learning: Distributional Robustness and Gradient Flows Yong Sheng Soh National University of Singapore
17:50-18:00	Closing remarks 10 min